Master's Degree in Data Science
Academic Year: 2025-2026

HOMEWORK 3

CUSTOMER CHURN PREDICTION - SPARK API

MARKETING ANALYTICS

PRESENTED BY : Matteo Fumagalli - id. 894468

DATASETS & PREPROCESSING

Dataset	Samples	Target Distribution
Dataset 1	40,000	52% churn - Balanced
Dataset 2	50,000	87% churn - Imbalanced

DATASET OVERVIEW & BUSINESS SCENARIOS

The analysis evaluates two distinct market scenarios across datasets sharing identical structures (13 features, 1 target variable):

- **Dataset 1** represents a **baseline scenario** in which roughly 1 out of 2 customers churns.
- **Dataset 2** captures a **high-risk business scenario**, where the majority of the customer base leaves.

TRAIN-TEST SPLIT

Datasets are split into training (75%) and test (25%) sets to ensure the evaluation of the model capability on unseen data. In particular, **stratified random sampling**, which preserves the exact churn/no-churn ratio across both training and test, is fundamental especially for the heavily skewed Dataset 2.

DATA PREPROCESSING

- **Imputation** steps are included into the pipeline for robustness, but both datasets do not have missing values.
- **Standard scaling** is applied to numeric variables to ensure comparable scales across variables.
- **StringIndexer**, followed by **OneHotEncoder**, is applied to categorical variables, producing sparse binary feature vectors suitable for machine learning models.

Features
Numeric: age, income, tenure_months, monthly_spend, website_visits, support_tickets, email_click_rate, loyalty_score, discount_use, last_campaign_days
Categorical: gender, region, campaign_type

After preprocessing, a **VectorAssembler** is used to combine all numerical and categorical feature representations into a single feature vector.

MODELS & HYPERPARAMETERS

Model	Hyperparameter	Values
Logistic Regression	regParam	[1, 2, 10, 100]
Random Forest	numTrees	[50, 100, 150]
	maxDepth	[5, 10, 15]
Gradient Boosting	maxIter	[50, 100, 150]
	stepSize	[0.05, 0.1, 0.2]

MODEL TRAINING & TUNING

Three machine learning models are selected: Logistic Regression, Random Forest, and Gradient Boosting. In particular, **hyperparameter tuning** is conducted using the **validation set** extracted from the training set to avoid data leakage. Moreover, to mitigate class imbalance, class weights are used to give more importance to the under-represented class. Once the optimal configurations are identified, models are **retrained on the full training set** and **evaluated on the test set**.

MODEL SELECTION

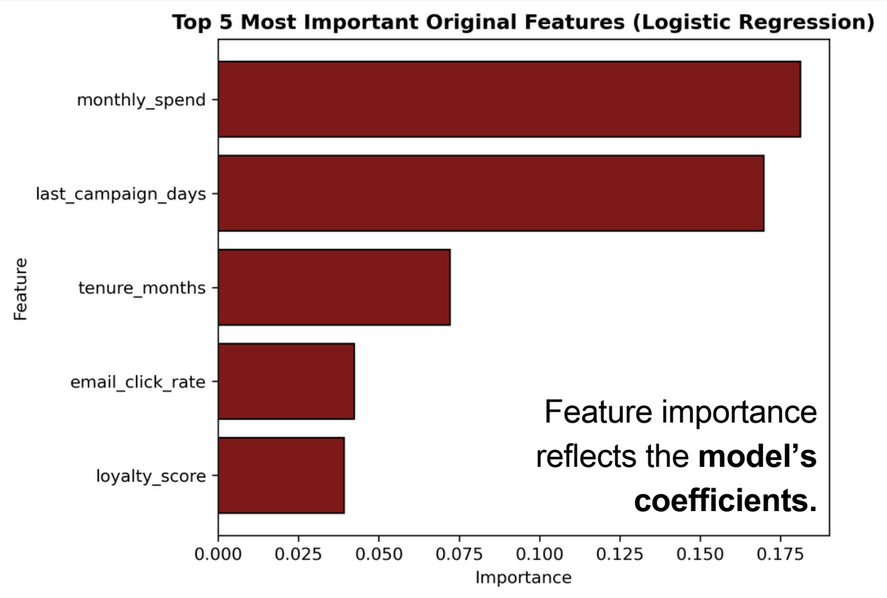
The **F1 score** is adopted as the primary measure for model selection. By balancing precision and recall, it addresses a critical trade-off between maximizing budget efficiency and capturing churning customers. In case of a tie, the model with the higher **ROC-AUC** score is selected. In particular:

- Dataset 1 - **Logistic Regression**: in a balanced market environment, a linear boundary proves highly effective, ensuring model interpretability and parsimony.
- Dataset 2 - **Random Forest**: as an ensemble of decision trees, it excels at capturing complex, non-linear patterns, while maintaining a balanced classification performance across both classes.

Dataset	Best Model	Hyperparameter	Value
Dataset 1	Logistic Regression	regParam	1
Dataset 2	Random Forest	numTrees	100
		maxDepth	15

EVALUATION & FEATURE IMPORTANCE

DATASET 1 (LOGISTIC REGRESSION)

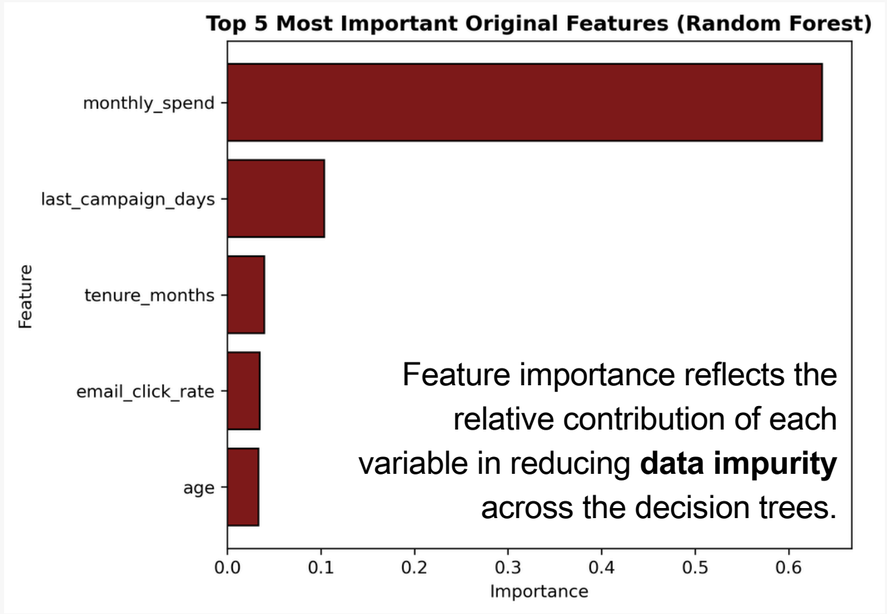


Confusion Matrix		Predicted	
		0	1
Actual	0 (no churn)	3,962	860
	1 (churn)	100	4,179

Metric	Value
Accuracy	0.81
F1 score	0.81
ROC-AUC	0.90

Logistic Regression shows balanced performance, being equally effective at predicting both churners and non-churners.

DATASET 2 (RANDOM FOREST)



Confusion Matrix		Predicted	
		0	1
Actual	0 (no churn)	1,201	390
	1 (churn)	386	10,524

Metric	Value
Accuracy	0.94
F1 score	0.94
ROC-AUC	0.97

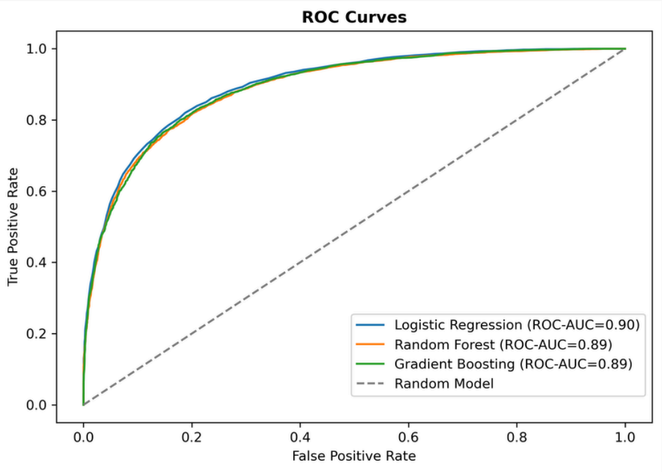
Random Forest achieves high overall metrics: it misclassifies about 25% of the non-churning minority and maintains strong churn detection performance.

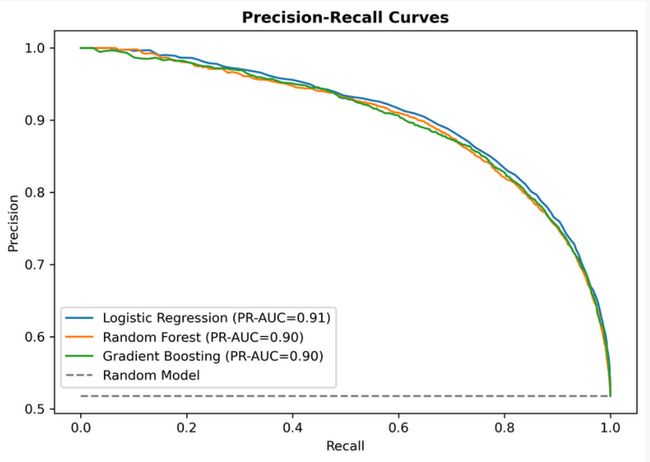
MULTI MODELS COMPARISON

Model comparison is based on evaluating the top-performing configuration of each of the three models for both datasets.

DATASET 1

	Linear Regression	Random Forest	Gradient Boosting
Accuracy	0.81	0.81	0.81
F1 score	0.81	0.81	0.81
ROC-AUC	0.90	0.89	0.89





DATASET 2

	Linear Regression	Random Forest	Gradient Boosting
Accuracy	0.85	0.94	0.91
F1 score	0.87	0.94	0.91
ROC-AUC	0.97	0.97	0.97

